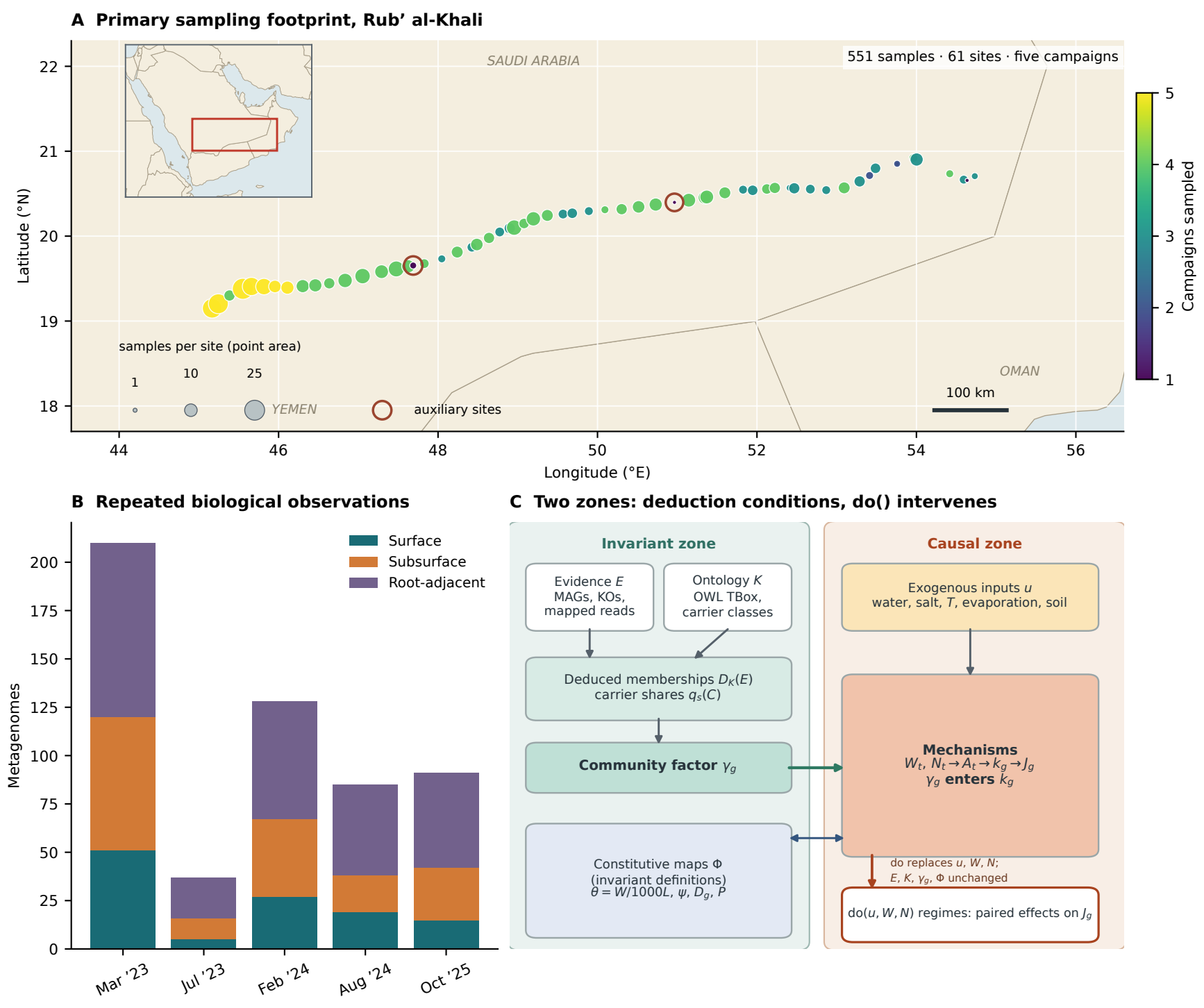

A Primary sampling footprint, Rub' al-Khali
551 samples · 61 sites · five campaigns
SAUDI ARABIA
YEMEN
OMAN
samples per site (point area)
1
10
25
auxiliary sites
100 km
Latitude (°N)
Longitude (°E)
Campaigns sampled
B Repeated biological observations
Surface
Subsurface
Root-adjacent
Metagenomes
Mar '23
Jul '23
Feb '24
Aug '24
Oct '25
C Two zones: deduction conditions, do() intervenes
Invariant zone
Evidence E
MAGs, KOs, mapped reads
Ontology K
OWL TBox, carrier classes
Deduced memberships $D_K(E)$
carrier shares $q_s(C)$
Community factor γ_g
Constitutive maps Φ
(invariant definitions)
$\theta = W/1000L$, ψ, D_g, P
Causal zone
Exogenous inputs u
water, salt, T, evaporation, soil
Mechanisms
W_t, $N_t \to A_t \to k_g \to J_g$
γ_g enters k_g
do replaces u, W, N;
E, K, γ_g, Φ unchanged
do(u, W, N) regimes: paired effects on J_g